

Original Article

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Reliability and thermal fatigue life prediction of solder joints for advanced automotive microelectronics

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Abstract With the development of the automotive industry, the number of automotive electronic components is growing rapidly. This leads to new requirements for reliability of the automotive electronics. Especially, for difficult operating environments, such as engine compartment of the car, extreme thermal cycles, and long holding time, where prediction of the fatigue life of the component is particularly important. In this study, thermal fatigue life prediction models for two packages are proposed: TSOP over BGA, and flip chip with BGA. The overall deformation of the two packages was analyzed using Moiré fringes, and was found to be consistent with simulation results. The reliability of the model was verified by comparing thermal shock results and simulation results. Then, the fatigue life of the component under thermal cycles was predicted using the model. The overall component life of the BGA with flip chip package was 818 cycles. For TSOP over BGA this was 1005 cycles, but the shortest life of the BGA was 725 cycles, which may result in the failure of the entire component. The fatigue life prediction using different models is helpful to assess and optimize components at the early stage of product development.

1. Introduction

System electronics are semiconductor devices, which perform processing functions such as handling digitized data or performing conversion. Unlike memory semiconductors, for system electronics operation efficiency can be achieved by integrating multiple functions into a single chip. System electronics are divided into PC, mobile, automotive, home and industrial appliances, depending on the application domain. By function, system electronics can be divided into microprocessors, logic or analog ICs (integrated circuits), complex digital ICs, and sensors. Today, the number of semiconductors in a car can be as much as 300, with about 2000 modules are expected to come in the next few years when autonomous vehicles will be commercialized [1].

Automotive semiconductors can be divided into three main categories: sensors, which detect environmental conditions, control devices, which determine the required behavior, and driving semiconductors, which perform certain actions. Also, with the advancement of the autonomous driving technology, in-vehicle networking (IVN) and in-vehicle infotainment (IVI) semiconductors are rapidly developing. Previously, most automotive semiconductors used packaging techniques such as lead grid arrays (LGA) or ball grid arrays (BGA). There is a growing number of flip chip packages, which meet quality requirements of the vehicle semiconductor market [1].

Flip chips and ball grid arrays (BGA) are composed of large number of pins and can simultaneously forward multiple electrical signals, thus their usage in system electronics is rapidly increasing [2]. This flip chip technology significantly reduces the area occupied by the package as compared to wire bonding with conventional marginal alignment. Also, external noise, electrical capacity, and inductance values are significantly smaller than in conventional packages that reduces electrical delay. However, flip chips inevitably generate heat because of electrical

resistance. As the packages become smaller, the resulting heat increases because of the reduction of the distributed area. Heat generation is sensitive to temperature and stress that affects the lifetime of the responding elements and can lead to system failures [3-5].

To obtain reliable surface-mounting of PCB assemblies, it is important to know how package bending changes with temperature. Component-to-component and component-to-PCB board surfaces shall remain flat relative one another. Failure to do so may result in surface defects such as head-in-pillow (HIP), opening, bridge, tensile, etc. The initial package flatness can be affected by various features of the component design and manufacturing. However, changes in flatness with temperature are mainly caused by the coefficient of thermal expansion (CTE) discrepancies between the different materials used in the package. Therefore, material CTE becomes an important factor in package design. There have been many studies of the reliability of the solder joints, temperature uniformity, and bending relationship of packages. Su et al. have studied the thermal cycling reliability of BGA packages with SAC305 and SnPb, IMC thickness and failure modes of the BGA components for different test vehicles [6, 7]. Hu et al. have proposed a thermal fatigue life prediction model for the BGA packages to take into consideration load sequence effects to decrease the difficulty of getting fatigue test data effectively [8]. Kwak used the thickness of the IMC layer and the diameter of the pump as variables to analyze temperature change, thermal stress, and thermal deformation performed [3]. Kim et al. have studied the fracture phenomenon of the IC package using the finite element method to determine the thermal characteristics according to the package structure, and the difference in the amount of deflection of the frame was confirmed when the lead frame of the TSOP (thin small outline package) was fixed and the pressure applied [9]. Moreover, Kwak and Chung have investigated not only an accurate methodology for forecasting the thermal fatigue life of the solder bumps jointed with Cu-pillar in the flip chip packages, but also the effect of the mechanical properties of the underfill materials [10]. However, especially BGA packages are widely used in the automotive across various systems, including engine control, brakes, communication, infotainment, and radar to name only a few. Under the hood applications typically require higher continuous working temperature and multiple heating-cooling cycles during the vehicle life. For example, the working temperature of the equipment should be up to 150 °C, and the component should feature 15 years of reliable operation with zero defect quality level. In addition, more and more equipment functions are integrated to meet new requirements for networked vehicles, autonomous driving, infotainment and sensor integration that increases the die size [11, 12]. To reduce the cost of the design and development of the automotive IC packages, meet the requirements, and shorten the time-to-market more reliable models are needed to predict the fatigue life and assess the reliability of the package.

In order to develop new materials and processes suitable for

future operations and shorten the development time of chip packaging, based on the previous studies, this paper attempts to propose a life prediction model for electronic packages that to predict the thermo-mechanical reliability of the overall component. First, solid-45 was used as the element type in FEA, and two IC package models were established: BGA with flip chip and TSOP over BGA for reference. To accurately simulate the fatigue behavior of solder interconnects induced by cyclic thermal loading, Anand constitutive model for the viscoplasticity of solder was used for numerical analysis. Second, to verify the FEA model, the warpage deformation measurement was carried out under the thermal loading and compared with FEA simulation results. Third, the fatigue life of the solder joint under cyclic thermal loading was predicted by using Darveaux's crack growth correlations and Weibull analysis [13]. As a result, the maximum characteristic life for considered electronic packages was determined as reliability estimation.

2. Numerical approach

The aim of this study is to use commercial finite element software (ANSYS 10.0) for predicting the deformation response in various stacked package designs. Because of the different rates of thermal expansion among the materials, warpage develops when temperature changes affect the package. These displacements lead to stresses and strains, which are the primary cause of low-cycle fatigue in the solder. To accurately simulate the displacements caused by the temperature load, three-dimensional models for different stacked package designs are used in this study. These models are developed using the quarter symmetry of the package. The symmetry we choose depends on various parameters like package configuration, die size, etc.

Through the previous study on the thermal fatigue analysis of the solder joint, solid-45 was used as the element type, and the viscoplastic materials of solder ball used Anand constitutive model for numerical analysis. With the concentrated plastic work result, The Darveaux's crack growth model was used to predict the thermal fatigue life of solder joint [14-17]. According to this method, parametric modeling and numerical analysis are carried out.

This study discusses parametric modeling of two package designs. First is the BGA with flip chip mounted underneath. Second, for the purpose of comparative analysis, we use a package-on-package structure with BGA mounted underneath of the TSOP. TSOP is the IC package using lead frame, which is a low-cost package for white electronics, automotive applications, desktop PCs, and so on. Its performance is not as good as that of the BGA or flip-chip CSP, but it is widely used because of the low price.

Fig. 1 shows cross-sectional details for design 1 package. Based on these details we build a parametric FE model shown in Fig. 2. We use model symmetry to save computational time without compromising the accuracy.

Fig. 3 schematically shows design 2: a BGA with a flip chip

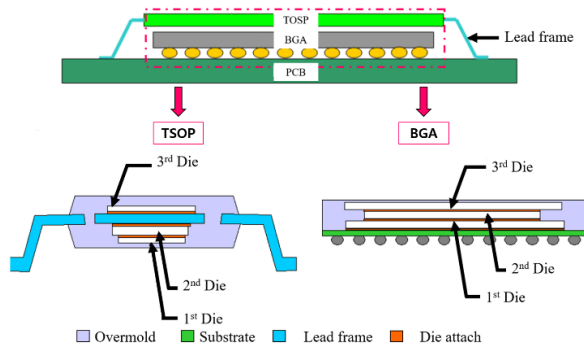


Fig. 1. TSOP over BGA package (design 1).

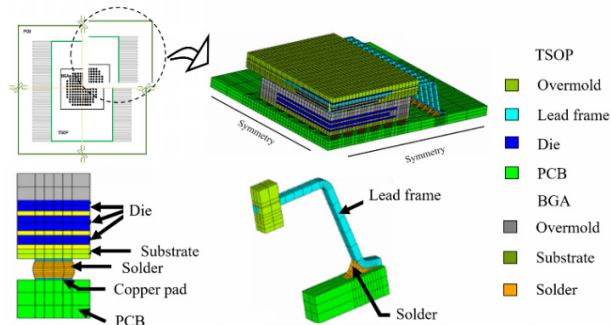


Fig. 2. FE parametric quarter symmetry model for TSOP over BGA package and details of the solder joint model.

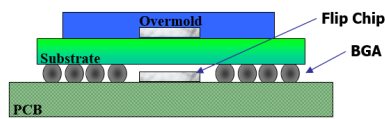


Fig. 3. BGA with flip chip (design 2).

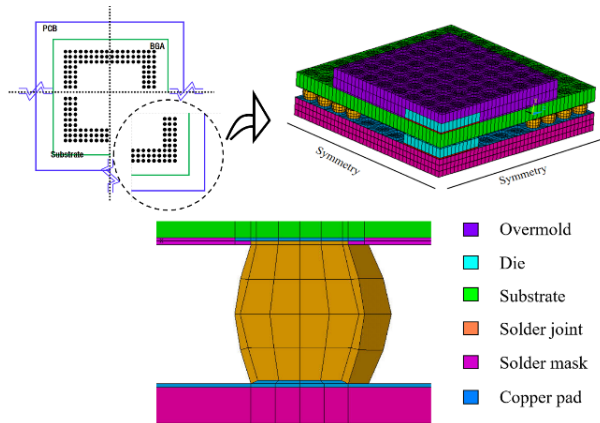


Fig. 4. 3D modeling of BGA with flip chip quarter symmetry model and details of the solder joint model.

mounted beneath. The base dimensions of the model are shown in Fig. 4. Model symmetry is used to build a quarter symmetry model for the package.

The essence of the parametric modeling is a parametric study. The user can observe the effect of various critical parameters on the desired output by changing a macro input file

Table 1. Dimensional details for designs 1 and 2.

Package	Parameters	Base model dimensions (mm)
Design 1: TSOP over BGA package	TSOP	Size
		14.5 x 13.2
		Overmold thickness
		0.78
		No. of leads
		32
		Leadframe pitch
		0.4
		Die size
		5 x 5
	BGA	Size
		10 x 11
		Solder ball pitch
		0.8
		Solder height
		0.28
		Substrate thickness
		0.13
		Die thickness
		0.46
Design 2: BGA with flip chip		Substrate size
		17 x 17
		Solder height
		0.63
		Pad diameter
		0.60
		Board thickness
		1.00
		Substrate thickness
		0.76
		Die size
		5 x 5
		Die thickness
		0.46

Table 2. Material characteristics for design 1.

Components	Elastic modulus E (Gpa)	Poisson's ratio	CTE (ppm/°C)
Silicon	168.92	0.36	2.6
Overmold (BGA)	20.68	0.3	14
Underfill	13.79	0.28	21
Cu (pad)	120.66	0.3	17
FR4 (PCB)	22.06	0.28	18
BT (substrate)	26.2	0.28	15
Solder	Temp. dependent (Table 4)		21
Lead frame (alloy 32)	150.30	0.3	5
Overmold (TSOP)	6.89	0.3	5

and rerunning the analysis. Table 1 outlines important geometrical parameters for both designs along with their base dimensions. Tables 2 and 3 show the Material attribution parameters for both designs. Table 4 shows the linear isotropic material properties of 63Sn37Pb eutectic solder.

To describe the relationships between the stress and the strain tensors, we can directly invoke a classical Anand constitutive model, which has been implemented using ANSYS. Anand model is widely used for solders with its accurate constitutive relationship, which can reflect deformation behavior of the viscoplastic materials related to the strain rate and the temperature, as well as the historical effect of the strain rate, strain hardening and dynamic recovery. Table 5 shows the Anand's constants for 63Sn37Pb [14-17].

Element type solid-45 was used for package modeling. This is a bilinear element defined by eight nodes with three degrees

Table 3. Material characteristics for design 2.

Components	Elastic modulus E (Gpa)	CTE (ppm/°C)
Overmold	55.16	15.00
Chip	172.4	3.50
Underfill	5.58	35.00
Substrate	55.16	15.00
Copper	128.93	15.00
FR-4	55.16	15.00

Table 4. Linear isotropic material properties of 63Sn37Pb eutectic solder.

	T1	T2	T3	T4	T5	T6	T7	T8
Temp (K)	193	208	218	273	298	338	378	398
Elastic modulus E (Gpa)	54.47	50.95	48.68	35.78	4.34	20.68	12.41	12.41
Poisson's ratio	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35

Table 5. Solder constitutive constants for ansys (Anand's constants).

Ansyz	Parameter	63Sn37Pb	Definition
C1	So (Mpa)	12.4	Initial value of deformation resistance
C2	Q/k (1/K)	9400	Activation energy / Boltzmann's constant
C3	A (1/sec)	4.0E+06	Pre-exponential factor
C4	ϵ	1.5	Multiplier of stress
C5	m	0.303	Strain rate sensitivity of stress
C6	ho (Mpa)	1379	Hardening constant
C7	S ^A (Mpa)	13.8	Coefficient for deformation resistance saturation value
C8	n	0.07	Deformation resistance value
C9	a	1.3	Strain rate sensitivity of hardening

of freedom at each node and translations in the nodal x, y, and z directions. The solders are modeled using Visco 107 elements. In this simulation we assume that:

- The package is stress free at the higher temperature (125 °C).
- The build-up of thermo-mechanical warpage is the result of the CTE mismatch among different package materials as the package cools down to final temperatures.
- Perfect adhesion exists at the material interfaces within the package.

3. Experimental validation

The CTE mismatch between the chip, the substrate, and the printed circuit board (PCB) results in thermally induced deformations within the electronic package assembly. Repeated thermal or power cycling builds up stresses in the interconnections and eventually leads to the device failure during operation.

Full field optical measurement techniques have been widely used to quantify deformations and the associated strains, and to locate risk areas of the electronic packages. Optical measurement techniques help obtain actual package behavior and thus can be used for experimental assessment of the reliability of different electronic and MEMS packages obtained using numerical simulation tools. These methods have a wide spectrum of application for material characterization and strain, and for deformation analyses. In this study, we attempt to measure the actual deformation of the sample packages under the thermal load, using a powerful optical measurement tool, namely Moiré interferometry.

3.1 Moiré interferometry (MI) technique

Moiré interferometry is based on the principle of interference of the light, whereby a diffraction grating replicated on the specimen interacts with the reference grating of the interferometer to produce the interference fringe pattern. The fringe pattern holds information about the displacement of the specimen. Depending on the wavelength of the light used, the grating frequency and the accompanying optics, the sensitivity of the interferometry system varies. The details of the Moiré interferometry measurement can be found in the previous studies [18-22].

When deformations must be recorded as a function of temperature, it is necessary to use a chamber, which provides heating and cooling functions. Some researchers have developed Moiré setups for real-time observations (Fig. 5) [23-25]. In this study the specimen was cross sectioned into halves as shown in Fig. 6, ensuring that the cross sectioned surface is perfectly flat. After the cross sectioned surface is polished flat, it is preheated to 90 °C and kept for about an hour until uniformly heated. A high-frequency (1200 lines/mm) crossline grating from an ultra-low expansion (ULE) grating mold is then replicated at 90 °C using a high-temperature curing epoxy and is kept until the epoxy cures. This temperature can be treated as the reference temperature or the undeformed state. When the specimen is brought back to the room temperature (25 °C), it's deformation grating features the deformation signature corresponding to the thermal load (DT = 65 °C).

The high frequency crossline diffraction grating applied to the specimen surface imitates the deformation of the specimen with high fidelity. With the interferometer, the ULE grating is used to tune a null field setting at room temperature, corresponding to the original uniform grating frequency.

Then, the interaction of the deformed specimen grating with the virtual reference grating of the interferometer creates the Moiré fringe pattern, which is observed with the imaging system. The contour interval between the fringes (without fringe shifting) is 0.417 μm per fringe order. From these fringe patterns, one can obtain deformation and strain. However, sometimes even such high resolution is not enough to provide accurate deformation data for small areas such as solder joint. The measured lengths are so small that for these regions the de-

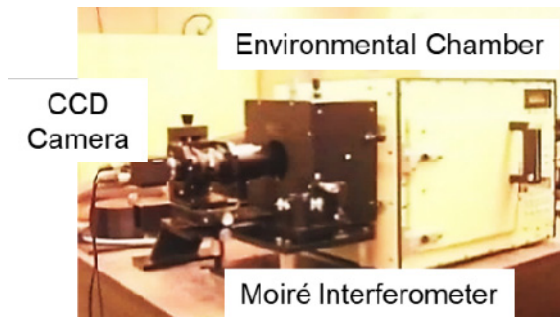


Fig. 5. Typical setup.

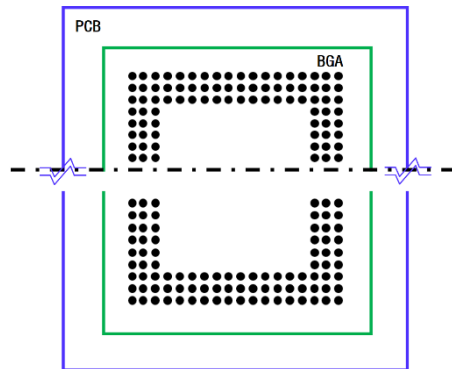


Fig. 6. Cross sectional Moiré.

formation must be interpolated. This deficiency can be addressed with the fringe shifting method, which uses multiple fringe patterns shifted by a certain amount to obtain fractional fringe orders that significantly increases the practical sensitivity of the fringe analysis. From the resulting data matrix of the calculation, a contour map can be plotted with a designated contour interval.

3.2 Comparison of FEA and experimental validation

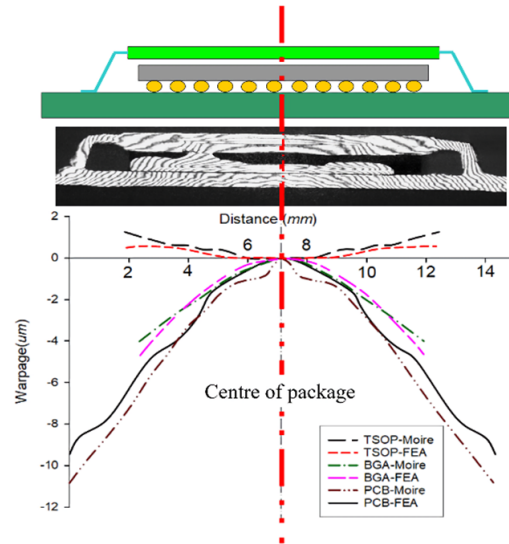
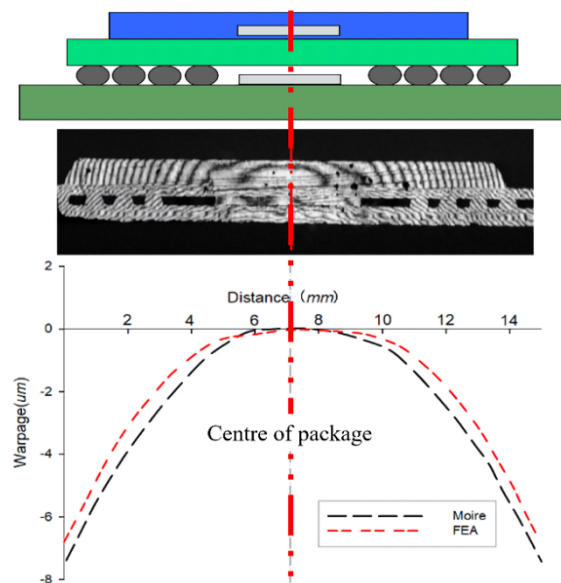
FE Model verification with Moiré interferometry results is done using the same 1/4 symmetry model we have built before. However, symmetry boundary conditions are applied to only one of the two edges where these conditions have been imposed before. This is because for Moiré interferometry analysis we slice the package into the halves.

For design 1, the TSOP, BGA and PCB modules are compared and analyzed, respectively. The results are shown in Fig. 7. The maximum deformation results for each module are shown in Table 6. One can see in the figure that the maximum deformation occurs on the PCB, which is the furthest from the neutral point. This is because the CTE of the PCB is higher than the effective CTE of the module, and the assembly warps down globally.

For design 2, the frequency of the specimen gratings is 1200 lines/mm. The contour interval between the fringes (without fringe shifting) is $0.417 \mu\text{m}$ per fringe order. The fringe pattern of the specimen after loading in vertical direction (denoted as V

Table 6. V field comparison of TSOP over BGA.

Component	Moiré $D_{\max}$ (μm)	FEA $D_{\max}$ (μm)
TSOP	1.2	0.6
BGA	4	5
PCB	10	9.5

Fig. 7. Result of Moiré fringe pattern V field (TSOP over BGA, $\Delta T = 65^\circ\text{C}$) & comparison of FEA.Fig. 8. Moiré fringe pattern V field (BGA with flip chip, $\Delta T = 65^\circ\text{C}$) & comparison of FEA.

field) is shown in Fig. 8. One can see from the figure that the maximum deformation of the FEA is $6.7 \mu\text{m}$, and the maximum deformation of the Moiré result is $7.4 \mu\text{m}$. This is the result of the CTE mismatch between the BGA module and the flip-chip mounted PCB. The most noteworthy result is the solder joint

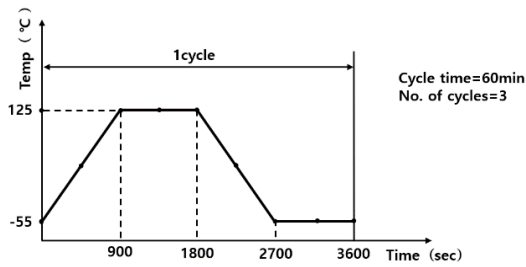


Fig. 9. Thermal cycling profile.

position with the maximum strain. The maximum strain occurs at the solder joint at the maximum distance from the neutral point. This is due to local and global CTE mismatches. Through the analysis of the design 2, one can see that simulation and experimental results are basically consistent.

In both the cases a close match between the FEA and Moiré results was observed. Thus, we could calibrate the FE parametric models for designs 1 and 2 with regards to experimental results from the Moiré interferometry, in assembly and individual (module only) configurations. Now, we can place sufficient confidence on the model with regards to its accuracy. The model can be assumed well-tuned to predict the fatigue life of the solder joints under thermal cycling conditions.

4. Fatigue life prediction and reliability analysis

In many cases, the failure of electronic equipment is caused by the fatigue of the solder joints. To predict the fatigue life of these elements, we use the combined finite element simulation and one-dimensional combined creep and time independent plastic model. This model helps find the number of cycles, which the component in question can withstand [26]. Darveaux and Perkins have proposed many fatigue life models. In this paper, the Darveaux's crack growth correlations is used [13].

4.1 Validated FEA model

By comparing the simulation results of the finite element model with the results of Moiré analysis, we determine the confidence of the established model. The FE model is subjected to thermal loads as shown below (Fig. 9). The first 900 s temperature rises from -55 degrees to 125 degrees Celsius, holding times are 900 s, and then the specimen cools during 900 seconds to -55 degrees with holding times 900 s, which makes a cycle. Holding time and temperature change rate are the same.

We simulate the package for 3 complete cycles. In this method, the strain energy density is averaged across the elements along the solder joint interface where the crack propagates. The average viscoplastic strain energy density accumulated per cycle is given by:

$$\Delta W_{ave} = \frac{\sum \Delta W^e \cdot V^e}{\sum V^e} \quad (1)$$

Table 7. Crack growth constants for 25.4 um layer.

K_1 (Cycles/MPa ² K ²)	K_2	K_3 (mm/cycle/MPa ⁴ K ⁴)	K_4
383	-1.62	1.93e-6	1.04

where ΔW_{ave} is volume averaged viscoplastic strain energy density, ΔW^e is viscoplastic strain energy density per cycle per interface element, and V^e is volume of each interface element.

In order to obtain a steady-state response, we need to simulate a sufficient number of cycles. Through the following equation, inelastic and total strain energy can be used to predict the package life with the help of empirical constants provided by Darveaux and Syed [13, 27, 28].

The characteristic life of the joint:

$$\alpha = N_0 + \frac{a}{da/dN} \quad (2)$$

where ΔW_{avg} is accumulated plastic work per cycle, $N_0 = K_1 (\Delta W_{avg})^{K_2}$ (no. of cycle to crack initiation), $\frac{da}{dN} = K_3 (\Delta W_{avg})^{K_4}$ (crack growth rate), and K_1, K_2, K_3, K_4 are crack growth constants.

We studied different number of layers: 12.7 um, 25.4 um and 38.1 um layer thicknesses. After convergence study and careful evaluation of the results, the 25.4 um thick layers near interfaces were selected for calculating the volume averaged plastic work per cycle. K_1, K_2, K_3 and K_4 are the Darveaux empirical constants shown in Table 7.

The studied packages can have from one to hundreds of individual joints, and a single failure can cause the whole component to fail. To calculate the overall package reliability, the reliability of each joint should be considered. If we assume that all solder joints are in series, we can use Weibull 2-parameter (2P) statistical distribution to express component reliability by following the series reliability equations:

Overall component reliability

$$R_c = \prod_{j=1}^m R_j \quad (3)$$

Reliability of the individual solder

$$R_j = e^{-q \left(\frac{N}{\alpha} \right)^\beta} \quad (4)$$

where β is the shape parameter, m is the number of solder joints in the series, q is the number of identical joints in the set 'j', α is the characteristic life, and N is the number of cycles to achieve the desired reliability.

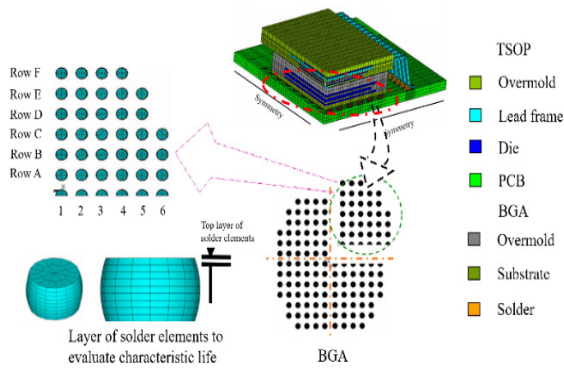


Fig. 10. Layer for evaluating volume averaged plastic work for BGA.

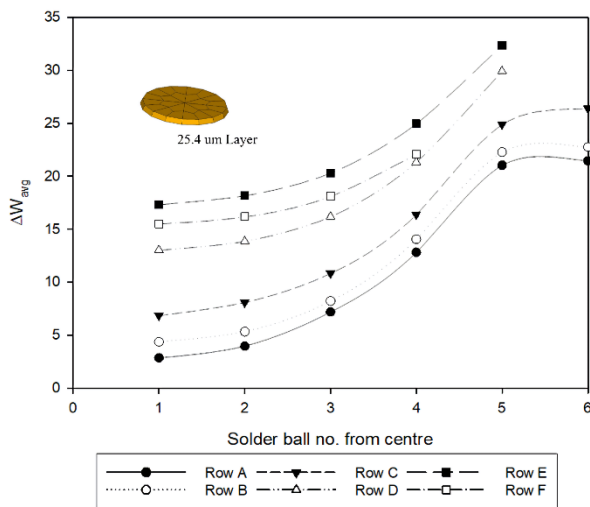


Fig. 11. Plastic work per cycle for BGA (small).

Fraction of population that fails in time 't'

$$F(t) = 1 - R(t). \quad (5)$$

4.2 Design 1: TSOP over BGA

4.2.1 BGA module

Using the simulation analysis, we found that the plastic work was the most concentrated near the package side. As shown in Fig. 10, the 25.4 um thick layer was taken to analyze the plastic work. The analysis of each solder according to its position is shown in Fig. 11. One can see that the closer the solder plastic work is to the edge, the more concentrated is the plastic work. Through Eqs. (1) and (2), the characteristic life of the solder at each position can be obtained, as shown in Fig. 12. From Figs. 11 and 12 one can see that the more concentrated plastic work is, the shorter is the characteristic life of solder joint. In Eq. (4), considering the occurrence of the early wear, when β is 4 and the percentage of the solder failure is 63.5 %, the characteristic lifetime of the overall component can be predicted as shown in Fig. 13, which shows that it takes 725 cy-

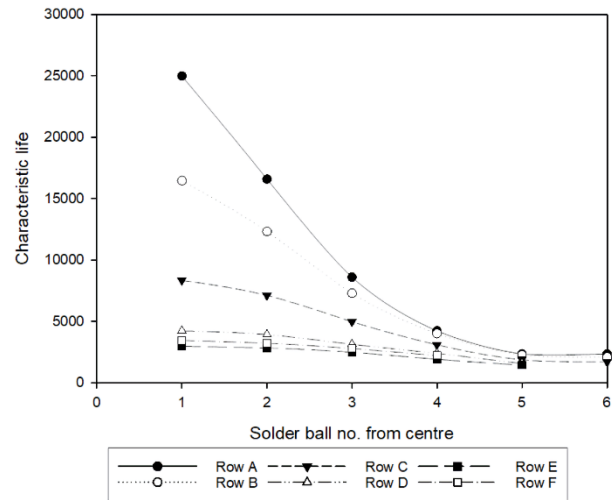


Fig. 12. Characteristic life for BGA (small).

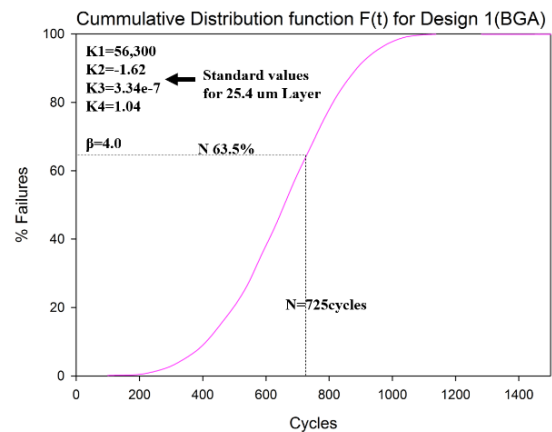


Fig. 13. Overall component life for BGA (small).

cles for 63.5 % failures.

4.2.2 TSOP

Like for the BGA, we choose a 25.4 um thick layer for the TSOP near the lead frame connection with the PCB. The volume averaged plastic work and consequently the characteristic life of each joint are calculated for all joints in the lead frames. Fig. 14 shows the plot of the characteristic life. It is obvious that the characteristic life decreases as we move away from the plane of symmetry. So, the joint farthest from the symmetry plane will have the minimal characteristic life. Considering the reliability and the probability of the failure of each joint, the component life of the TSOP can be plotted as the percentage of failures against the corresponding life. Fig. 14 shows the N63.5 life of the TSOP for 1005 cycles.

4.3 Design 2: BGA with flip chip

Also, we can plot the component life for the BGA (big) as shown in Fig. 15, which shows the N63.5 life of the BGA for 818 cycles.

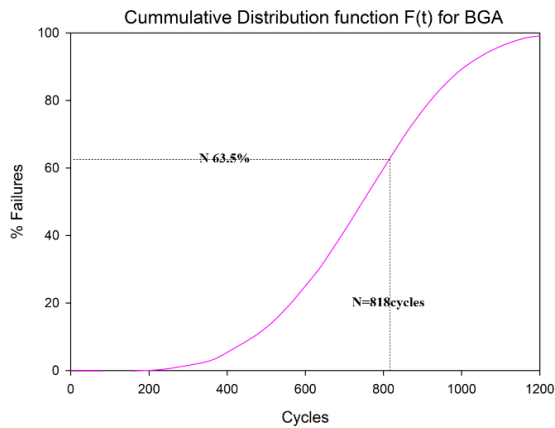


Fig. 14. Overall component life for the TSOP.

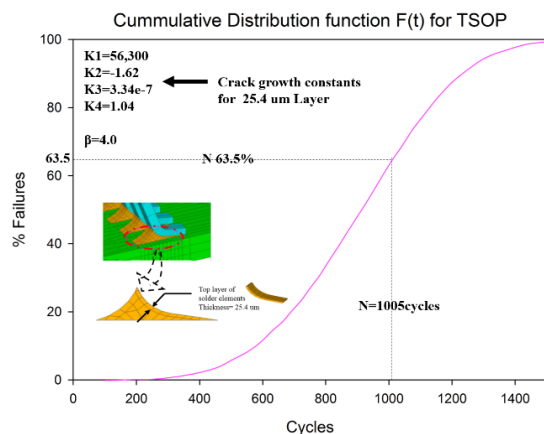


Fig. 15. Overall component life of the BGA with flip chip.

5. Conclusions

In order to minimize the cost of packaging design and reduce the pressure of listing time, two packaging models are proposed to predict the fatigue life. Through the experimental and simulation analysis of the proposed models under thermal cycle conditions, the following conclusions can be drawn:

1) Through the analysis of the Moiré fringes of the two packages under thermal shock conditions, it was found that due to the difference of the module CTE, the package produces deformation as a whole, with main deformation concentrated in the solder joint, which is furthest away from the neutral point. The agreement between the experimental results and the simulation results fully confirms the accuracy of the model and prediction results correlate well with the experimental data.

2) Under the same thermal mechanical conditions, the overall component life of the design 1 (TSOP over BGA) is longer than that of the design 2 (BGA with flip chip), and its reliability is higher. However, for the BGA and TSOP modules of design 1, the shortest life of the BGA is 725 cycles, which may result in the failure of the entire component.

3) Using this fatigue life prediction model, in addition to the thermal shock, the fatigue life of the components can be further studied in terms of the stress and the strain under different

loads such as vibration that can help reduce the difficulty of obtaining fatigue data and minimize the error. It is helpful to design evaluation and optimization at the early stage of product development.

Although currently most solder materials for automotive electronics are Pb-contained solders, replacing them with Pb-free solders is expected following the RoHS (restriction of the use of certain hazardous substances) requirements. Therefore, future works may consider new solder materials to carry out reliability prediction method suggested in this study.

Acknowledgments

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Nomenclature

LGA	: Lead grid array
BGA	: Ball grid array
CTE	: Coefficient of thermal expansion
TSOP	: Thin small outline package
ΔW_{ave}	: Volume averaged viscoplastic strain energy density
ΔW^e	: Viscoplastic strain energy density per cycle per interface element
V^e	: Volume of each interface element
ΔW_{avg}	: Accumulated plastic work per cycle

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